

Tutorial

1. Explain whether the following function can be a solution to TISE: (a) Ax , (b) $A/\sin(x)$, (c) A (d) $A \exp(-x^2)$
2. Prove that the time independent wave function can always be taken to be real.
3. Prove that in 1-D, there are no degenerate bound state.
4. If $V(x)$ is an even function, then show that $\psi(x)$ can be taken to be either even or odd.
5. Show that the wavefunctions for TISE corresponding to two different potential $V_1(x)$ and $V_2(x)$, differing by more than a constant cannot be the same.
6. Show that the energy E must exceed the minimum value of $V(x)$ for every normalizable solution of time independent Schrodinger equation (TISE).